LAB4: Trajectory Optimization

Individual Deliverables 1 and 2

Submission account: `yawangjiang`

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1 Deliverable 1: Single-Segment Optimization

The problem is to minimize $J = \int_0^1 [P'(t)]^2\,dt$, with $P(0) = 0$ and $P(1) = 1$. Write $P(t) = \sum_{i=1}^n p_i t^i$, so that $p_0 = 0$ is eliminated. Then

$$J = \mathbf{p}^\top Q_n \mathbf{p}, \qquad (Q_n)_{ij} = \frac{ij}{i+j-1}, \quad 1 \le i, j \le n. \tag{1}$$

The objective uses no factor $1/2$; a solver using $\frac{1}{2}\mathbf{p}^\top H \mathbf{p}$ should receive $H = 2Q_n$.

1.1 1. Degree One

For $P(t) = p_1 t$, the final-position constraint gives $p_1 = 1$. Thus

$$\boxed{P(t) = t, \qquad J = 1.} \tag{2}$$

1.2 2. Degree Two

For $\mathbf{p} = [p_1, p_2]^\top$,

$$Q_2 = \begin{bmatrix} 1 & 1 \\ 1 & 4/3 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad b = 1. \tag{3}$$

In particular, $J = p_1^2 + 2p_1p_2 + \frac{4}{3}p_2^2$. Substituting $p_1 = 1 - p_2$ yields $J = 1 + \frac{1}{3}p_2^2$. Therefore

$$\boxed{\mathbf{p}^* = [1, 0]^\top, \qquad P(t) = t, \qquad J^* = 1.} \tag{4}$$

Allowing the quadratic term does not reduce the optimum.

1.3 3. Degree Three and the Euler-Lagrange Result

For $\mathbf{p} = [p_1, p_2, p_3]^\top$,

$$Q_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4/3 & 3/2 \\ 1 & 3/2 & 9/5 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \qquad b = 1. \tag{5}$$

The minimizer is $\mathbf{p}^* = [1, 0, 0]^\top$, again giving $P(t) = t$ and $J^* = 1$. Indeed, Cauchy-Schwarz gives

$$1 = \left(\int_0^1 P'(t)\,dt \right)^2 \le \int_0^1 [P'(t)]^2\,dt. \tag{6}$$

Equality holds for the constant derivative $P' = 1$. This agrees with the Euler-Lagrange equation $P'' = 0$ and the two position boundary conditions.

1.4 4. Add the Terminal-Velocity Constraint $P'(1) = -2$

The coefficient relation for this additional condition is $\sum_{i=1}^{n} ip_i = -2$. For each degree, solve

$$\min_{\mathbf{p}} \mathbf{p}^\top Q_n \mathbf{p} \quad \text{subject to} \quad A\mathbf{p} = \mathbf{b}. \tag{7}$$

One direct method is the equality-constrained KKT system

$$\begin{bmatrix} 2Q_n & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}. \tag{8}$$

4(a). Degree Two

Use the same Q_2 and

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{9}$$

The constraints give $p_1 + p_2 = 1$ and $p_1 + 2p_2 = -2$, hence

$$\boxed{P(t) = 4t - 3t^2, \qquad J = \int_0^1 (4 - 6t)^2 \, dt = 4.} \tag{10}$$

There is only one feasible quadratic coefficient vector.

4(b). Degree Three

Use Q_3 and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{11}$$

Solving the KKT equations gives

$$\mathbf{p}^* = \begin{bmatrix} 1/4 \\ 9/2 \\ -15/4 \end{bmatrix}, \qquad \boxed{P(t) = \frac{1}{4}t + \frac{9}{2}t^2 - \frac{15}{4}t^3, \qquad J^* = \frac{17}{8}.} \tag{12}$$

The boundary values satisfy $P(0) = 0$, $P(1) = 1$, and $P'(1) = \frac{1}{4} + 9 - \frac{45}{4} = -2$. Its cost is $17/8 = 2.125 < 4$, so the additional cubic freedom improves the constrained quadratic solution.

Summary of Deliverable 1

Allowed degree and constraints	Optimal polynomial	Cost
1, 2, or 3; endpoint positions only	t	1
2; also $P'(1) = -2$	$4t - 3t^2$	4
3; also $P'(1) = -2$	$\frac{1}{4}t + \frac{9}{2}t^2 - \frac{15}{4}t^3$	17/8

2 Deliverable 2: Multi-Segment Optimization

Assume fixed, positive segment durations, specified waypoint positions, fixed derivatives through order $r-1$ at the two outer endpoints, and free derivatives of orders $1,\dots,r-1$ at the internal waypoints. Counts below are per scalar coordinate. Independent spatial coordinates repeat the same calculation.

2.1 1. Minimum Snap: $r = 4$ and k Segments

For $J = \sum_{i=1}^{k} \int [P_i^{(4)}(t)]^2 \, dt$, the Euler-Lagrange equation is $P_i^{(8)} = 0$. Each segment therefore has degree at most 7 and 8 coefficients, giving $8k$ unknown coefficients.

Count the equations using the following disjoint categories:

Category	Equations	Meaning
Waypoint positions	$2k$	Two endpoint position values for each segment, representing $k+1$ distinct geometric waypoints.
Fixed nonzero-order derivatives	6	Velocity, acceleration, and jerk at the overall start and finish.
Free internal derivatives	$6(k-1)$	Each of the three free derivatives at every internal waypoint yields continuity and a natural optimality equation.

At an internal waypoint, freedom in derivatives $j = 1, 2, 3$ gives, respectively, continuity of derivative orders $(1, 6)$, $(2, 5)$, and $(3, 4)$. The second member of each pair is the natural stationarity condition obtained by integration by parts. Thus

$$2k + 6 + 6(k-1) = 8k. \tag{13}$$

Equivalently, classify the internal positions separately: $2(k-1)$ internal position equations, 8 fixed endpoint conditions of orders 0 through 3, and $6(k-1)$ free-derivative equations. Endpoint positions must not be counted twice.

2.2 2. General Derivative Order r

The Euler-Lagrange equation is $P_i^{(2r)} = 0$, so each segment has degree at most $2r-1$ and $2r$ coefficients, for $2rk$ unknowns. Internal free derivative j gives continuity of order j and the natural matching condition of order $2r-1-j$. Consequently

$$\boxed{2k + 2(r-1) + 2(r-1)(k-1) = 2rk.} \tag{14}$$

The three terms count waypoint positions, fixed outer derivatives of orders $1,\dots,r-1$, and internal free-derivative continuity/optimality equations. The equivalent endpoint-inclusive count is $2(k-1) + 2r + 2(r-1)(k-1) = 2rk$.

For $r = 4, k = 3$, this gives $6 + 6 + 12 = 24$ equations for 24 coefficients. A free derivative is not an extra prescribed value: its equations follow from continuity and optimality. The highest derivative $2r-1$ may jump at a fixed internal position waypoint.

Source

MIT VNAV LAB4 exercise statement: `https://vnav.mit.edu/labs_2023/lab4/exercises.html`.